

Kubernetes task - 2

EKS cluster create with name kds-eks

The screenshot shows the Amazon Elastic Kubernetes Service (EKS) console. The cluster 'kds-eks' is in the 'Node group creation in progress' state. A blue banner at the top indicates that the node group is being created and may take several minutes. Below the banner, there are buttons for 'Connect', 'Actions', and 'Monitor cluster'. A warning message states: 'End of standard support for Kubernetes version 1.33 is July 29, 2026.' The 'Cluster info' section shows the following details:

Cluster info	Kubernetes version	Support period	Provider
Status: Active	1.33	Standard support until July 29, 2026	EKS
Cluster health: 0	Upgrade insights: 0	Node health issues: 0	Capability issues: 0

Node group created with 2 nodes

The screenshot shows the Amazon Elastic Kubernetes Service (EKS) console. The cluster 'kds-eks' is now in the 'Nodes (2)' state. The 'Node groups (1)' section shows the following details:

Group name	Desired size	AMI release version	Launch template	Status
kd-nodegroup	2	1.33.13-20260625	-	Active

Created 3 EC2 instances

The screenshot shows the Amazon EC2 console. The 'Instances (2)' section shows the following details:

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...
eks-cluster-ser1	i-0a9a37a1bb1638f50	Running	t3.medium	3/3 checks passed	ap-south-1c	ec2-13-200-19-41.ap-s...	13.200.19.41
eks-cluster-ser2	i-0fcc3aa70a792bd9e	Running	t3.medium	3/3 checks passed	ap-south-1a	ec2-13-126-35-57.ap-s...	13.126.35.57

The screenshot shows the Amazon EC2 console. The 'Instances (1/5)' section shows the following details:

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...
eks-cluster-ser1	i-0a9a37a1bb1638f50	Running	t3.medium	3/3 checks passed	ap-south-1c	ec2-13-200-19-41.ap-s...	13.200.19.41
eks-cluster-ser2	i-0fcc3aa70a792bd9e	Running	t3.medium	3/3 checks passed	ap-south-1a	ec2-13-126-35-57.ap-s...	13.126.35.57
member-server	i-069739fd65efbadbd	Running	t2.micro	2/2 checks passed	ap-south-1b	ec2-13-233-74-176.ap-...	13.233.74.176
member-server	i-06b1cb59a587c5ee6	Running	t2.micro	2/2 checks passed	ap-south-1b	ec2-13-201-22-51.ap-s...	13.201.22.51
root-server	i-0024cd0b6fa711169	Running	t2.micro	2/2 checks passed	ap-south-1b	ec2-13-201-76-208.ap-...	13.201.76.208

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aws [Search] [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) kousicko (7863523052)
[ec2-user@ip-172-31-5-135 ~]$ aws eks update-kubeconfig --name kds-eks
Added new context arn:aws:eks:ap-south-1:786352305260:cluster/kds-eks to /home/ec2-user/.kube/config
[ec2-user@ip-172-31-5-135 ~]$ kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-172-31-21-188.ap-south-1.compute.internal Ready    <none>   48m    v1.33.13-eks-7d6f6ec
ip-172-31-45-36.ap-south-1.compute.internal Ready    <none>   48m    v1.33.13-eks-7d6f6ec
[ec2-user@ip-172-31-5-135 ~]$
[ec2-user@ip-172-31-5-135 ~]$ kubectl create deployment nginx --image=nginx
deployment.apps/nginx created
[ec2-user@ip-172-31-5-135 ~]$ kubectl get deployment
NAME    READY    UP-TO-DATE    AVAILABLE    AGE
nginx   1/1      1             1            12s
[ec2-user@ip-172-31-5-135 ~]$
[ec2-user@ip-172-31-5-135 ~]$ kubectl expose deployment nginx --port=80 --type=LoadBalancer
service/nginx exposed
[ec2-user@ip-172-31-5-135 ~]$
[ec2-user@ip-172-31-5-135 ~]$ kubectl get svc
NAME    TYPE    CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
kubernetes ClusterIP  10.100.0.1    <none>         443/TCP     62m
nginx   LoadBalancer  10.100.51.37  ab40426705e64447ea5c63648c9a0f8b-373872820.ap-south-1.elb.amazonaws.com 80:30361/TCP 11s
[ec2-user@ip-172-31-5-135 ~]$
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aws [Search] [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) kousicko (7863523052)
[ec2-user@ip-172-31-5-135 ~]$ kubectl get all
NAME                                READY    STATUS    RESTARTS    AGE
pod/nginx-5869d7778c-86f95         1/1      Running   0            7m40s

NAME    TYPE    CLUSTER-IP    EXTERNAL-IP    PORT(S)
AGE
service/kubernetes ClusterIP  10.100.0.1    <none>         443/TCP
68m
service/nginx     LoadBalancer  10.100.51.37  ab40426705e64447ea5c63648c9a0f8b-373872820.ap-south-1.elb.amazonaws.com 80:30361/TCP
6m44s

NAME    READY    UP-TO-DATE    AVAILABLE    AGE
deployment.apps/nginx  1/1      1             1            7m40s

NAME    DESIRED    CURRENT    READY    AGE
replicaset.apps/nginx-5869d7778c  1          1          1        7m40s
[ec2-user@ip-172-31-5-135 ~]$
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